

Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification

Hunmin Lee[✉], Ming Jiang[✉], Jinhui Yang[✉], Zhi Yang[✉], and Qi Zhao[✉]

Abstract—In recent years, significant strides in deep learning have propelled the advancement of electromyography (EMG)-based upper-limb gesture recognition systems, yielding notable successes across a spectrum of domains, including rehabilitation, orthopedics, robotics, and human-computer interaction. Despite these achievements, prevailing methodologies often overlook the intrinsic physical configurations and interconnectivity of multi-channel sensory inputs, resulting in a failure to adequately capture relational information embedded within the connections of deployed EMG sensor network topology. This oversight poses a significant challenge, impeding the extraction of crucial features from collaborative multi-channel EMG inputs and subsequently constraining model performance, generalizability, and interpretability. To address these limitations, we introduce novel graph structures meticulously crafted to encapsulate the spatial proximity of distributed EMG sensors and the temporal adjacency of EMG signals. Harnessing these tailored graph structures, we present Graph Convolution Network (GCN)-based classification models adept at effectively extracting and aggregating key features associated with various gestures. Our methodology exhibits remarkable efficacy, achieving state-of-the-art performance across five publicly available datasets, thus underscoring its prowess in gesture recognition tasks. Furthermore, our approach provides interpretable insights into muscular activation patterns, thereby reaffirming the practical effectiveness of our GCN model. Moreover, we show the effectiveness of our graph-based input structure and GCN-based classifier in maintaining high accuracy even with reduced sensor configurations, suggesting their potential for seamless integration into AI-powered rehabilitation strategies utilizing EMG-based gesture classification systems.

Index Terms—Explainable AI, generalizability, graph neural network, graph representation, hand gesture classification, surface electromyography.

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I. INTRODUCTION

APPROXIMATELY three million individuals across the globe face the challenges of upper limb amputation [1]. This disability makes it more difficult to engage in daily activities, leading to a substantial reduction in their overall quality of life [2]. As a result, surface electromyography (sEMG) has emerged as a viable technique, enabling the non-invasive recording of neuromuscular activation from residual muscles. This recording technique serves as a crucial interface for prosthetic control, offering the potential to bridge the gap between the amputee's intended actions and the prosthetic (predicted) actions.

The neurological topology in an amputee's forearm experiences substantial changes due to the loss of the limb and its associated muscles. Notably, muscles within the forearm that demand intricate dexterity control undergo significant restructuring during the rehabilitation process, as documented in [3], [4], [5], and [6]. This reconfiguration leads to considerable variation among individuals. Despite the availability of EMG-based prosthetic hands in the market, these devices are often limited to basic grip patterns and cannot be generalized across various individuals and environments. This limitation leads to a significant gap between the potential capabilities of these prosthetics and the intuitive, dexterous control required to fully restore an amputee's functional hand.

The goal of EMG-based gesture classification is to classify the specific gestures linked to a group of time-series EMG signals that display spatial and temporal patterns, indicating the muscle activities involved in the gesture. Motor units in the nervous system and associated muscular groups collaborate in different patterns to execute different coordinated movements [7], [8]. Therefore, modeling and understanding common activation patterns of muscles is crucial for accurate gesture classification [9]. To achieve this goal, previous studies have explored features within the upper limb's muscle signals using advanced machine-learning models. Many of them leverage convolutional neural networks (CNNs) [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27] and investigate the spatial correlations embedded within image representations. Specifically, conventional CNN models learn features from raw EMG signals, such as the temporal and frequency domain

features [10], [14], [15], [16], [19], to reveal the latent patterns present in the EMG signals. Once these features are extracted, they are transformed into an image based on the sequence of feature extraction and sensor index. State-of-the-art CNN-based methods can classify over 50 different gestures with over 90% accuracy on public benchmarks [11], [12]. Despite the promising performance of CNNs, muscle activations during different gestures involve intricate patterns that can be challenging to capture with CNN-based approaches. To further improve the classification accuracy, recent studies have proposed the use of graph neural networks (GNNs) [28], [29], [30], particularly graph convolution networks (GCNs) [29], [31], [32], motivated by their ability to capture the complex dependencies present in EMG data. These methods incorporate techniques like graph convolution, attention mechanisms, and dynamic graph structures to enhance gesture classification accuracy.

Current EMG-based gesture classification methods have demonstrated promising results [33], [34] by learning each user's specific gesture patterns. However, without considering the physical configurations of EMG sensors, existing methods face significant limitations in generalizability and interpretability: First, they cannot handle the variability of EMG data across different users or datasets. This limitation arises due to the influence of several external factors, including inter-subject variability, muscle fatigue, and shifts of the sensors, making it challenging for existing CNN and GNN methods to adapt effectively. Second, these methods often struggle to provide a clear and interpretable visualization of the connections between the signals recorded from various muscle groups and the gesture labels. This lack of interpretability hinders not only the practical application of EMG-based gesture classification but also our ability to gain insights into the underlying physiological processes to explain why certain sensors are more informative for distinguishing gestures. Finally, existing GNN or GCN methods only represent the spatial EMG features as a graph, without considering the dependencies of EMG signals across temporal windows, which limits their effectiveness in classifying gestures with complex temporal patterns.

Aiming at generalizable and interpretable EMG-based prosthetic control, we propose a novel GCN approach that uncovers the complex spatiotemporal patterns across multiple EMG sensors and provides interpretability to identify significant muscular regions that contribute to the classification of gestures. Different from previous GNN studies that implicitly learn a graph structure from the training data, we explicitly construct graph structures that mirror the spatial proximity of the EMG sensors in the physical world, while also representing the temporal adjacency of EMG signals. This enables the GNN classifier to better aggregate the innate spatiotemporal interactions across distributed EMG sensors and signals.

To illustrate this idea, Fig. 1 shows the positions of the 16×8 sensors based on the configuration of the CapgMyo dataset [12], as well as the sensor activations across time windows. First, as can be observed from the spatial distributions of EMG sensors, physically proximate sensors tend to share similar levels of activations, which suggests the importance of analyzing signals according to the spatial positions of the

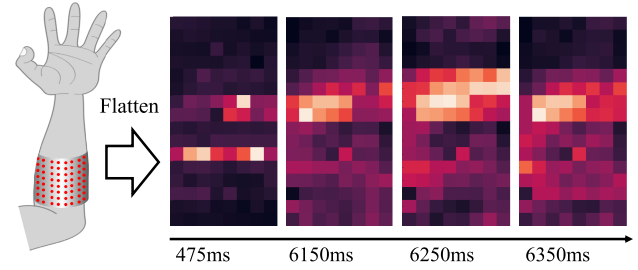


Fig. 1. We employ GCN-based classifiers to recognize gestures based on the spatiotemporal graphs. The heatmaps show the sensor activation patterns across different time windows: proximate sensors exhibit similar activations, emphasizing spatial analysis, while activation patterns across adjacent time windows highlight temporal relationships. To enhance the generalizability and interpretability of gesture classification, we propose explicitly structured graph representations that integrate the spatial proximity of EMG sensors and the temporal adjacency of EMG signals. This empowers classifiers to gain deeper insights into spatiotemporal correlations among EMG sensors and signals.

sensors. Second, it can be seen that the activation patterns across adjacent time windows (*i.e.*, 6150 ms, 6250 ms, and 6350 ms) are similar, which highlights the necessity to model the temporal relationship for EMG-based gesture classification. Therefore, to characterize the two types of patterns, we propose to combine spatial proximity and temporal adjacency with explicitly structured graph representations.

Using explicitly designed graph structures in GNN models has a key advantage: it enables the models to learn meaningful and interpretable EMG patterns that are robust against inter-subject variance. Our approach equips classifiers with a deeper understanding of the underlying data dynamics, independent of individual subjects. This allows for better generalization, even when tested on new subjects, ensuring robust performance across diverse scenarios. Through comprehensive experiments, we compare the performances between distinct graph structures across five publicly available datasets. Our results demonstrate the consistent superiority of GCN models based on these graph structures, outperforming state-of-the-art methods on multiple EMG datasets. Moreover, our method goes beyond classification accuracy and generalizability; it opens a window into gesture classification and provides transparent and intuitive visualizations that naturally explain the subtleties of gesture classification. This transparency is important for the advancement of prosthetics and robotics, as it provides researchers and practitioners with invaluable insights into how these models make decisions. Further, our method can outperform state-of-the-art methods with a significantly reduced number of sEMG sensors, demonstrating its efficiency and practicality. These achievements position our method at the forefront of EMG-based gesture classification, promising advancements with profound implications for prosthetics and robotics.

In summary, this paper introduces several significant contributions that collectively advance the field:

- 1) We propose novel graph structures that jointly capture spatiotemporal features based on the spatial proximity of distributed EMG sensors and the temporal adjacency of EMG signals. This unique approach differentiates itself

from existing methods that learn a graph model purely from data.

- 2) Building upon the constructed graph structures, we develop GCN gesture classifiers. These classifiers demonstrate significant performance gains on multiple datasets with different network architectures. In particular, they maintain high accuracy even when tested on distinct subjects excluded from the training data, showcasing their strong generalizability.
- 3) Our graph structures also enable the interpretation of the learned classifiers. Centering around the significance of muscular activation, our analyses demonstrate coherent patterns observed across subjects and gestures, contributing to a deeper understanding of the relationships between EMG signals and gestures.
- 4) Finally, our extensive experimentation on reduced sets of sensors demonstrates improved model efficiency and practicality of our approach, highlighting its potential impacts on effective sensor utilization in real-world applications.

The following sections of this paper are structured as follows. In Section II, we discuss related EMG gesture classification studies, including CNN-based methods and GNN-based methods. Section III introduces the proposed graph structures and feature representations that effectively aggregate spatiotemporal information. This section also introduces a novel GCN model and methodologies used to attain interpretability through the designed model. The experimental details and comprehensive analysis of results are presented in Section IV. Finally, we conclude the paper in Section V by summarizing our work and providing further insights.

II. RELATED WORKS

In this section, we review previous studies on deep learning methods to classify hand gestures based on EMG signals. Our literature review covers conventional CNN-based methods and the recent development of GNN-based models for EMG classification.

A. CNN-Based Approaches

In the past decade, various neural network architectures and learning methods have been developed to classify hand gestures using EMG data. The most prevalent designs are based on convolutional neural networks [10], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [26], [27], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], which effectively learn spatial features in a high-dimensional feature space. The pioneering work by Atzori et al. [13] extracted time domain (TD) and power-spectrum domain (PSD) features on the window-frame segments from the raw EMG signal and conducted gesture classification, which has been accepted as a standard procedure in EMG-based hand gesture classification [33], [34]. Inspired by this, various CNN-based classifiers [12], [13], [14], [21], [22], [23], [24], [25], [26] and feature-engineering techniques [52], [53], [54], [55], [56], [57] have been developed to effectively extract spatial features from EMG signals. Moreover, novel learning

paradigms have been imported to design a reactive EMG signal classification pipeline to achieve higher classification performance, such as transfer learning [40], [41], [42], [43], reinforcement learning [37], [38], [39], [50], attention-based model [17], [48], unsupervised learning [44], [45], [58], temporal convolution-based learning [14], [15], [46], [47], [49], multi-view learning [19], and few-shot learning [35]. Most of the studies successfully classify more than 50 gesture types with high accuracy performance when training and testing on the same subject.

Despite the significant achievements observed in gesture classification accuracy, conventional CNN-based models overlook the spatiotemporal relevance inherent in concurrent muscle regions and their significance during the execution of specific gestures. As a result, many of these methods are designed to target high intra-subject performance in very limited settings, without considering the generalizability of models. In particular, due to substantial inter-subject variability, these models struggle to exhibit consistent performance across different individuals. Moreover, these models frequently lack interpretable explanations for their predictions. In contrast, our innovative approach addresses these challenges by utilizing graph structures that capture both the spatial positions of EMG sensors and the temporal dynamics of signals. This not only significantly improves intra-subject and inter-subject classification accuracies but also provides intuitive and transparent explanations for our model's predictions, establishing our method as a comprehensive and forward-thinking solution in the field of gesture classification.

B. GNN-Based Approaches

Recently, methodologies centered around GNN models for EMG-based gesture classification have been introduced [28], [29], [30], [31], [32]. These approaches present strategies for efficiently extracting features derived from convolution. Most of these studies have integrated embedded locational encoding information [29], [30], [32] to extract spatial features, thereby processing this information within graph convolutional layers. Some also exploit recurrent neural network-based layers [30] and temporal convolution layers [28], [31] to encode temporal features. Specifically, STCN-GR [31] introduced a dynamic graph structure integrated with input EMG vectors. This architecture incorporated graph convolutional layers and temporal convolutional layers to extract spatiotemporal features. A variation of this method, GTGR-Net [28], enhanced classification performance by employing graph attention networks in an additive manner. Massa et al. [29] devised an independent pipeline for feature extraction based on high-density surface EMG (HD-sEMG) data. They established an intuitive nearest-neighbor graph structure, employing GraphSage layers within a GNN framework to derive spatial features and perform gesture classification. With a similar graph structure, Lee et al. [30] introduced a self-attention-based GNN approach that combined LSTM and GCN modules. Furthermore, Yang et al. [32] curated EMG and EEG datasets, combining information from these distinct input sources through the creation of a multimodal graph, capturing features from both modalities, facilitating accurate gesture classification.

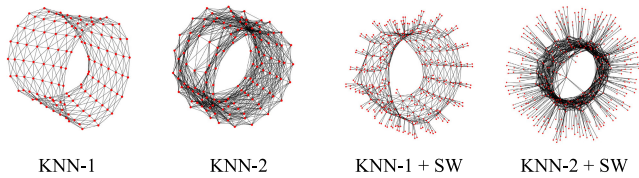


Fig. 2. Our approach introduces a novel method of constructing graph structures, enabling a physiologically meaningful representation of spatial and temporal relationships among the sensors. These examples illustrate our proposed graph structures tailored for the CapgMyo [12] dataset, showing varying degrees of connections, including KNN-1, KNN-2, and sensor-wise connections. Note that the FC and FC+SW graphs in CapgMyo are not visualized due to the high density of connections among the 16×8 sensors.

While GNN-based methods in principle allow models to capture the complex relationships within graph inputs, previous studies learn graph models implicitly from EMG data. Due to the significant variability of EMG datasets, these methods cannot generalize well across diverse dataset categories or subjects because they lack the knowledge of sensor configurations. Their implicitly learned models also lack biological interpretability, preventing the understanding of muscular activities and corresponding EMG signals that contribute to specific gestures. Furthermore, prior models fail to consider temporal connections within the graph topology. Although some studies incorporate temporal modules [28], [30], [31], their approaches are inherently limited as they process spatial and temporal information with separate modules. Our GCN model is different from existing methods by explicitly incorporating spatial and temporal dependencies in the graph structure, which allows the classifier to jointly learn the spatiotemporal patterns of EMG data. Furthermore, we introduce an interpretive methodology that quantitatively evaluates the contributions of individual EMG sensors, which allows us to analyze the corresponding muscular region's significance in classifying different gestures.

III. METHODOLOGY

To facilitate EMG-based gesture classification, prior studies have constructed high-quality datasets, each with various arrays of surface-attached sensors and distinct gesture classes [11], [12], [15]. Nevertheless, these datasets exhibit different sensor configurations and gesture classes. Given that many previous methods are often tailored to a specific dataset under intra-subject evaluation, their generalizability across different datasets, subjects, and network architectures is limited. It is also difficult to interpret these networks' predictions because of the lack of connections between the signals recorded from various muscle groups and the gesture labels. To address the crucial issues of generalizability and interpretability, our approach revolves around constructing graph structures that faithfully represent the sensors' physical configurations within each dataset. To address the crucial issues of generalizability and interpretability, our approach revolves around constructing graph structures that faithfully represent the sensors' physical configurations within each dataset and incorporates these structural graphs with EMG signals to train our GCN model for gesture classification. As shown in Fig. 2, these graph structures are designed to preserve the

spatiotemporal features and mirror the spatial relationships found within the physical arrangement of EMG sensors. This approach adeptly captures the intricate spatial configurations embedded in biological systems, thereby facilitating iterative model refinement driven by these interconnections. Based on the graph structures, we further present the GCN model for gesture classification, as well as model interpretation methods by quantifying the contributions of each EMG sensor to each gesture class of each individual (see Fig. 5).

A. Problem Definition

The complexity inherent in human biological anatomy operates within a substantially high-dimensional framework [59]. This highlights the imperative of considering a multitude of relational factors existent across biosystems. Similarly, human gestures exhibit complex patterns and interactions across diverse anatomical regions, where coordinated movements spanning joints and muscles are intricately intertwined, each component exerting influence on others while fulfilling specific tasks [60], [61], [62]. To realize this concept, we configure the arbitrary EMG signal sets to formulate spatial relationships across localized EMG signals, constructing a topology that enables deep learning models to distill pertinent information reflective of anatomical activities for robust gesture recognition. Despite the advancements in deep learning-driven gesture recognition studies, the existing models are often limited to considering arbitrary relationships across localized muscles and their activities. The CNN-based gesture recognition models, for instance, tend to arbitrarily align the electrode features and train them to a classifier, limiting the potential of spatial information across individual electrodes that cover unique localized muscular regions. Consequently, this inadvertently disregards nuanced correlations across distributed electrodes and their diverse features, resulting in a constrained biological feature space during the feature extraction process. To address this limitation, we propose a graph-based representation that mirrors the electrode configuration. This representation facilitates the establishment of interconnections among participant electrodes and their features, thereby fostering the dissemination of crucial information for robust gesture classification.

Based on the graph representation, we leverage the GCN-based gesture classification model. The core motivation behind this approach is to empower the GCN model to robustly capture both global and local relationships among all data channels, utilizing graph topologies that enable capturing salient spatiotemporal features. By doing so, we aim to develop models that are not only capable of achieving state-of-the-art performance but are also inherently adaptable and generalizable across diverse datasets and scenarios. This emphasis on learning generalizable models is important in extending the utility of EMG-based gesture classification beyond controlled environments, addressing the critical need for real-world applicability.

B. Graph Representations

While window-based CNNs have proven effective in EMG classification [13], [17], [19], [20], [24], they fall short in

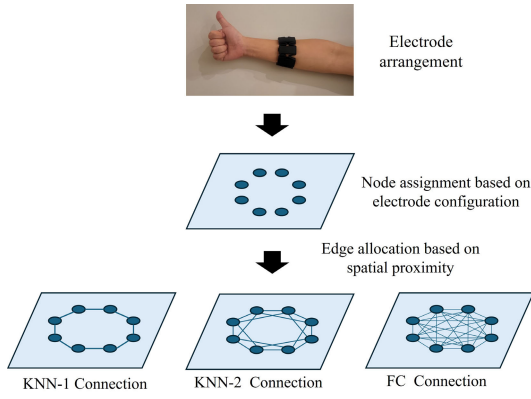


Fig. 3. Overview of constructing a spatial graph topology based on the physical configurations of multi-electrodes.

capturing relationships among distributed sensors [29], [30], [32]. Existing GNN-based models have aimed to learn these relationships by treating EMG sensors as nodes in graph models. However, they have overlooked the spatial configurations of EMG sensors and the temporal relationships of EMG signals. In this section, we propose, for the first time, a novel spatiotemporal graph representation that uses undirected graphs for both spatial and temporal aspects. The undirected nature of the graph allows for symmetric information flow between nodes, capturing the reciprocal interactions of muscles more accurately than directed graphs. This enables the model to learn both localized spatial dependencies and dynamic temporal patterns within the context of the specific dataset, providing a more comprehensive understanding of the spatiotemporal characteristics of EMG signals during gesture execution.

1) Spatial Graph Construction: The complexity of hand motion gestures arises from the fact that they involve a wide array of muscular activities working in concert. Each gesture is not simply the result of one muscle group contracting but a coordinated effort involving multiple muscles. These muscles are spread throughout the hand and forearm, and their precise activation patterns vary depending on the specific gesture being performed. To address such complexity, our spatial graph structures are constructed according to the proximity of sensors based on their positions, where sensors are considered graph nodes, and the graph edges are determined by the spatial proximity between sensors in the physical world. Fig. 3 provides a visual overview of the construction of spatial graph representations based on the physical arrangement of sEMG electrodes used for signal acquisition. As shown in the figure, our method focuses on two types of spatial topologies: K-nearest neighbors (KNN) and fully connected (FC) structures.

Specifically, based on the 2D electrode configuration map $\mathbf{V}$ utilized for the signal measurement, we construct the graph structure. The matrix $\mathbf{V}$ represents a sensor channel (*i.e.*, vertex) configuration with dimensions $M \times N$, where each element v_i denotes a sensor channel, as indicated in Equation (1). Let $G = \{\mathbf{E}, \mathbf{V}, \Omega\}$ denote a graph, where $\mathbf{E}, \mathbf{V}, \Omega$ signify a set of edges, vertices, and edge weights, respectively, with corresponding elements represented as e, v ,

and ω . For instance, $e_{(i,j)} \in E$ denotes the edge connecting nodes v_i and v_j . In the context of the CapgMyo dataset, M is allocated as 8 and N as 16, resulting in a CapgMyo electrode configuration characterized by an 8×16 sensor map.

$$\mathbf{V} = \begin{bmatrix} v_1 & v_2 & \cdots & v_N \\ v_{N+1} & v_{N+2} & \cdots & v_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ v_{(M-1)N+1} & v_{(M-1)N+2} & \cdots & v_{MN} \end{bmatrix} \quad (1)$$

In the graph representation, nodes are interconnected based on defined spatial relationships with their adjacent neighbors. The connectivity patterns of first-nearest-neighbor graphs (KNN-1) are expressed as follows:

$$e_{(i,j)} = \begin{cases} 1 & \text{if } d(i, j) \leq \sqrt{2} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where 1 represents a connection between the assigned vertices, and $d(i, j)$ indicates the Euclidean distance between the electrodes i and j in the array. The distance between an arbitrary node and its adjacent neighbors along the horizontal and vertical axes in an $M \times N$ matrix is set to 1. The distinctions within the indices determine the connection boundary, where the constraints limit traversal to a single adjacency as defined in the conditions. The second-nearest-neighbor (KNN-2) graph structure is similar, with the index boundary determining edge assignment extended from $\sqrt{2}$ to $2\sqrt{2}$:

$$e_{(i,j)} = \begin{cases} 1 & \text{if } d(i, j) \leq 2\sqrt{2} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Additionally, FC graphs exhibit maximum density, embodying a topology wherein every node is linked to every other node except itself, as elucidated in Equation (4) where L represents the larger value between M and N . This configuration ensures that all nodes constitute a set of neighbors reachable within a single-hop distance, with the scope of this notation adapting according to the spatial attributes governing sensor displacement. This high-density connection design fosters efficient information aggregation within each dataset.

$$e_{(i,j)} = \begin{cases} 1 & \text{if } i \neq j \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

2) Spatiotemporal Graph Construction: As demonstrated in prior studies [15], [28], [31], [50], [51], [54], temporal domain features are highly important to EMG-based gesture classification. To capture these dynamics, we introduce sensor-wise (SW) temporal connections, linking corresponding sensors across consecutive time windows.

We construct the spatiotemporal graph by first creating a separate spatial graph G_S (KNN or FC) for each time window. These G_S graphs capture the spatial relationships within each window. Crucially, the SW connections then link corresponding nodes (representing the same sensor) across consecutive spatial graphs G_S . This forms the overall spatiotemporal graph G_{SW} (see Fig. 4). Mathematically, an edge exists between nodes $i = (t, r)$ and $j = (t', r')$ if and only if they represent

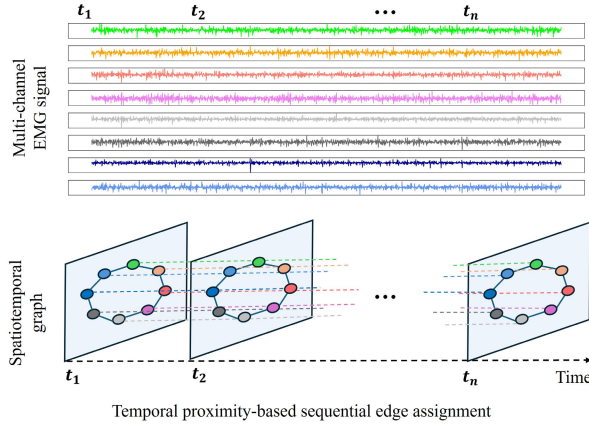


Fig. 4. Overview of constructing a spatiotemporal graph topology by temporally aligning spatial graphs and connecting identical node types across time steps.

the same sensor ($r = r'$) at consecutive time steps ($t' = t \pm 1$):

$$e_{((t,r),(t',r'))} = \begin{cases} 1 & \text{if } (t' = t \pm 1 \text{ and } r = r') \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

Unlike previous GNN methods that prioritize spatial relationships, our approach integrates both spatial and temporal information within a unified graph structure, enabling more effective learning of dynamic EMG patterns. This combined approach allows our model to capture both the intra-window spatial dependencies and the inter-window temporal dynamics. For example, as shown in Fig. 2, KNN-1+SW uses a KNN-1 spatial graph for each window, linked temporally by SW connections. This flexible framework can be adapted to different datasets and sensor configurations, resulting in a comprehensive and dataset-specific spatiotemporal representation.

C. GCN-Based Gesture Classification

GCNs [63], [64] stand as a specialized architecture of GNN, designed to tackle data organized in the form of graphs. The intrinsic strength of GCNs lies in their ability to uncover intricate relationships embedded within the graph structure by efficiently aggregating information from neighboring nodes via convolution process, often driven by spatial or temporal correlations. In this section, we introduce a GCN-based gesture classification approach that uniquely leverages our explicitly defined graph topology. Our GCN-based gesture classification model adopts the Message-Passing Neural Network (MPNN) architecture, facilitating the information exchange among neighboring nodes within the graph, thereby enabling the aggregation of spatiotemporal features to generate refined representations. Our model utilizes a standard GCN architecture [65], which performs message passing based on the normalized Laplacian matrix. This allows us to effectively capture both spatial and temporal relationships in the gesture data. Our approach centers on a node classification task, where each node corresponds to an electrode. In our framework, each node is allocated with a unique gesture label and its features, with multiple nodes participating in the same gesture sharing

identical labels. Given that each node shares similar information with others involved in the same gesture, the process of classifying individual nodes remains consistent throughout the entire graph within that specific gesture. Collectively, these nodes capture essential relational and contextual features from their interconnected counterparts, enhancing the model's ability to discern and classify various gesture types.

1) *Model Process*: Let $\mathbf{W}$ denote a parameter of our GCN model, and $\mathbf{X}$ represent the set of node features, where $\mathbf{x}_i$ denotes the feature vector of node i . Utilizing $\mathbf{x}_i$, we encode these features using Equation (6) with the encoding function $g(\cdot)$.

$$g(\mathbf{X}, \mathbf{W}_1, \mathbf{b}_1) := \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \rightarrow \mathbf{h} \quad (6)$$

Here, $\sigma(\cdot)$ represents non-linear activation function, and the weight matrix $\mathbf{W}_1 \in \mathbb{R}^{(n(\mathbf{x}) \times m)}$ and bias term $\mathbf{b}_1$ transforms the input feature vector $\mathbf{x}$ into refined feature representation $\mathbf{h} \in \mathbb{R}^{(m \times 1)}$. In each node v_i , we aggregate the $\mathbf{h}_j$ and create a message m per node, as illustrated in Equation (7) where l denotes layer, c_i is a normalization factor, v_j represents a neighbor node of node v_i , such that $N(v_i) \ni v_j$. This process is similar to the *pooling* process, summing up the neighbor features to emphasize discernible features.

$$m_{v_i}^{(l+1)} = \sum_{v_j \in N(v_i)} \frac{1}{c_i} \mathbf{h}_{v_j}^l \quad (7)$$

The combined message is subsequently concatenated with the embedded features of v_i . This concatenated vector is then normalized using the L2 Norm, yielding a node embedding vector $\mathbf{N}$ expressed as follows:

$$\mathbf{N}_i = \frac{\hat{m}}{\sqrt{\sum_j m_j}} \text{ where } \hat{m} = (\mathbf{h}_{v_i}^{(l+1)} || m_{v_i}^{(l+1)}) \ni m_j \quad (8)$$

where $||$ denotes concatenation operation. We then iterate through equations (6), (7), and (8), represented collectively as $f(\cdot)$. This expands the receptive field during the integration process, leading to generating effective representation for extracting coarse-grained features. Additionally, we utilize a fully connected layer ($g(\cdot)$) to derive prediction $\hat{y}$, expressed as follows:

$$\hat{y}_i = \sigma(g(f(\mathbf{N}_i, i), \mathbf{W}, \mathbf{b})) \quad (9)$$

where index i is required to iteratively call the neighbor nodes via an adjacency matrix. Leveraging the prediction $\hat{y}$, we compare the ground-truth y and apply the categorical cross-entropy loss function $L(\mathbf{y}, \hat{\mathbf{y}}_t)$ to compute the difference derived from $\mathbf{W}$ and then aggregate these differences over all samples i and classes j at epoch t .

$$L(\mathbf{y}, \hat{\mathbf{y}}_t) = -\frac{1}{n(\mathbf{X})} \sum_{\forall i} \sum_{\forall j} y_{ij} \log(\hat{y}_{ij}) \quad (10)$$

Therefore, we summarize our objective function as follows:

$$\lim_{t \rightarrow T} \arg \min_{\mathbf{W}} L(\mathbf{y}, \hat{\mathbf{y}}_t) \quad (11)$$

Our graph convolution-based model is designed to extract and capture the spatial features encoded within the graph

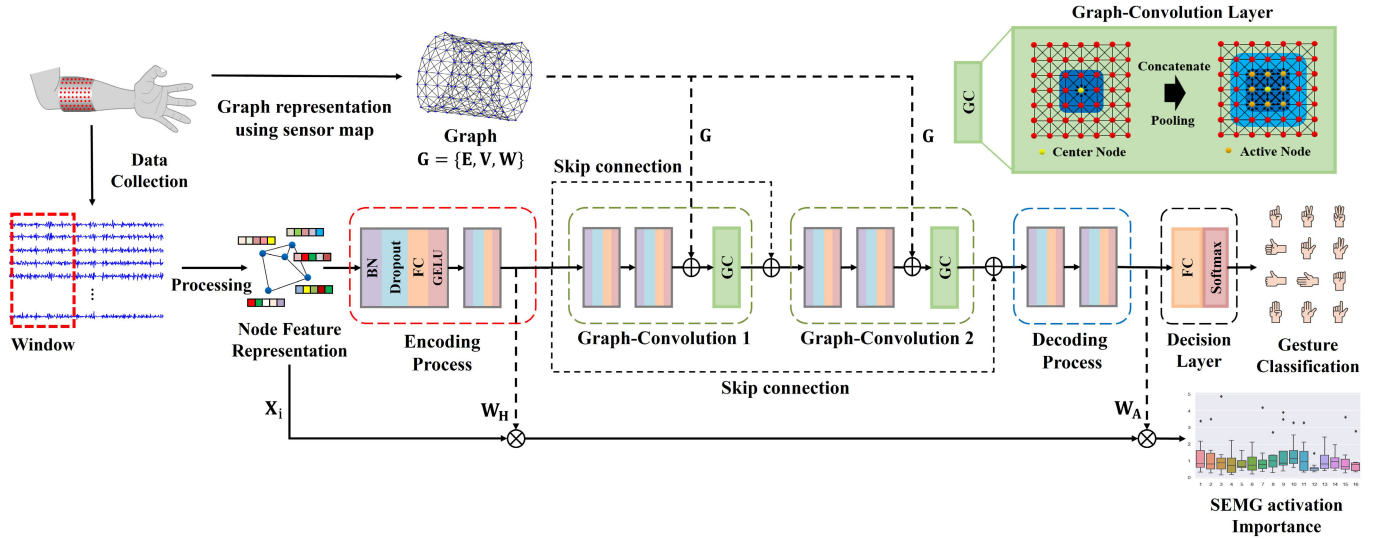


Fig. 5. Overview of the proposed gesture classification and model interpretation methods. Based on defined graph structures, we develop a GCN method to aggregate key attributes across multiple sensors and their encoded features for gesture classification. In addition, to interpret the contributions of each sensor to the classification, we propose an integrated analysis based on the learned model weights and data from various sensors.

structure. In the EMG features measured from electrodes, where signals from muscle activity are often represented as time-series data or spatially distributed signals, our model can capture dependencies between different channels or time points utilizing tailored graphical structure inputs, thereby effectively exploiting the inherent structure in the data for gesture classification tasks.

2) Model Architecture: Details of the proposed GCN-based gesture classification model architecture are depicted in Fig. 5. Here, the model takes a batch of nodes as input, processing their node features with an encoding layer, two graph convolution layers, and a decoding layer. The encoding and decoding layers are composed of an identical structure, with BN-Dropout-FC layers stacked twice, where BN represents batch normalization, and FC denotes a fully connected layer. In our model, we standardized the configuration of FC layers by employing a layer size of 256 nodes, utilizing the Gaussian Error Linear Unit (GELU) activation function. We maintained uniform dropout rates of 0.2 across all FC layers, except for an initial dropout layer where the rate was set to 0. This approach ensured that no information loss occurred from the initial inputs during the network's training.

Our architecture begins with the encoding layer, which initializes node features by generating feature embedding vectors, transforming raw input data into localized representations within each node. It also performs feature normalization, aligning node features to facilitate smooth and efficient information propagation across the graph. This step ensures that the model can effectively learn and update node representations during the convolutional process, capturing and aggregating meaningful patterns. Following the encoding layer, our model incorporates two graph convolution (GC) layers based on the GCN architecture. This implementation follows the spectral graph convolution approach [65], where the layers perform localized filtering on the graph structure by leveraging the

normalized Laplacian matrix. Each GC layer is followed by two fully connected (FC) layers, which contribute to refining the feature representations learned during the graph convolution process. These FC layers further enable non-linear transformations of the node features, enhancing the model's ability to capture complex gestural patterns and ensuring that the model captures both local neighborhood interactions and global structural information. Additionally, we employ a skip connection which allows the encoded output to bypass the GC layers to serve as a form of regularization to reduce overfitting. This becomes particularly important when the input graph exhibits dense connections, such as those created using KNN-2+SW and FC+SW. Furthermore, the graph representation described in Section III-B is fed into the GC layer, facilitating the propagation of refined gestural features along the network paths that reflect the topology of the sEMG sensor configuration. This process enables the dissemination of relevant biological information across nodes by dynamically updating their features through neighborhood aggregation, where each node's representation is iteratively refined by integrating information from its neighboring nodes. Following this, the merged features proceed to the decoding layer. Here, the features are renormalized and refined, ensuring consistency and stability before being passed to the final decision layer. In the decision layer, logits are computed to determine the final output of gesture prediction.

D. Model Interpretation

While deep learning models have exhibited remarkable performance, their inherent complexity often renders them black-box methods, lacking the transparency and interpretability necessary for real-world applications. This opacity poses a significant barrier to understanding how these models arrive at their decisions, hindering their practical utility. Recognizing this critical issue, we introduce a novel interpretation approach

that leverages the parameters extracted from the fully connected (FC) layers of our GCN model. By doing so, we enable a deeper level of insight into the classifier's decision-making process. This interpretative framework provides a clear and comprehensible understanding of how the model relates to EMG signals and gestures.

As shown in Fig. 5, we perform a matrix multiplication operation between the input data samples, denoted as $\mathbf{X}_i \in \mathbb{R}^{S \times \tau}$, and the weights $\mathbf{W}_H \in \mathbb{R}^{\tau \times P}$ from the initial FC layer. In this context, S refers to the number of input sensors, τ is the number of node features, and P is the layer size. This procedure extracts distinctive features unique to individual gestures. Next, we multiply the output with weight matrix $\mathbf{W}_A \in \mathbb{R}^{P \times G}$ of the final layer, with G indicating the overall number of gesture classes for classification. The outcome, denoted as $\mathbf{R}$, has a dimension of $(S \times G)$, which conveys the level of activation for each sensor and each gesture, as mathematically shown in Equation (12).

$$\mathbf{R}_i = \mathbf{X}_i \cdot \mathbf{W}_H \cdot \mathbf{W}_A \quad (12)$$

Subsequently, we compute the average of the absolute values of $\mathbf{R}_i$ from all samples for a given trial ($\forall i$) and then across all trials ($\forall t$) for each gesture class (g) and subject (s), which leads to the final output of distributed sensor importance denoted as $\mathbf{I}_{(s,g)}$:

$$\mathbf{I}_{(\exists s, \exists g)} = \frac{1}{n(\forall t) \times n(\forall i)} \sum_{\forall t} \sum_{\forall i} |\mathbf{R}_{(t,i)}| \quad (13)$$

The above method uses data from various sensors and the learned model weights to analyze the contributions of each sensor to the classification of gestures, and the aggregation result of Equation (13) offers the potential to discern the relevance of muscular regions and their activities in the classification of distinct gestures.

IV. EXPERIMENTS

In this section, we present comprehensive experiments to demonstrate the effectiveness of the proposed approach. First, we present experimental details, including datasets, training, and evaluation details. Second, we evaluate the gesture classification performance of our model by comparing it to state-of-the-art methods. We demonstrate the generalizability of our method across different datasets and network architectures, using both intra-subject and inter-subject evaluation approaches. Finally, we visualize the gesture classification results using our proposed interpretation methodology to gain further insights from the experimental results.

A. Experimental Settings

1) **Datasets:** Our experimental investigations were conducted on five well-established public EMG datasets, including Ninapro DB5 [11], CapgMyo Type A and Type B [12], UCI EMG Pattern Database [66], and BandMyo [15]:

- 1) The objective of the Ninapro project is to facilitate research on advanced hand myoelectric prosthetics by providing publicly available datasets. In our experiment,

we utilize Database 5, which encompasses EMG signals captured from 16 synchronized EMG sensors and encompasses 53 distinct hand gesture classes, including a rest class. The gesture taxonomy within DB5 comprises three main categories: basic finger movements (gestures 1-12), hand configurations and wrist movements (gestures 13-30), and grasping and functional movements (gestures 31-52), with detailed descriptions provided in [11]. Data acquisition involved 6 repetitions performed by 10 intact subjects.

- 2) The CapgMyo Type A and Type B datasets feature 128 high-density sensors and incorporate data from 18 and 20 intact subjects, respectively. These datasets encompass 8 isometric and isotonic gesture action classes, each comprising 10 repetitive trials. Preprocessing of the CapgMyo dataset involved band-pass filtering within the 20–380 Hz range, 16-bit A/D conversion, and normalization to the range of $-1 \sim 1$, corresponding to voltage values of $-2.5\text{mV} \sim 2.5\text{mV}$. The gesture class aligns with the first 8 gestures in Ninapro DB5 Exercise B, encompassing Thumb up (1), Extension of index and middle, flexion of the others (2), Flexion of ring and little finger, extension of the others (3), Thumb opposing base of little finger (4), Abduction of all fingers (5), Fingers flexed together in fist (6), Pointing index (7), Abduction of extended fingers (8).
- 3) The UCI EMG Pattern Database incorporates data from 36 subjects, employing 8 EMG sensors. This dataset covers 7 specific gesture classes, which are hand at rest (1), hand clenched in a fist (2), wrist flexion (3), wrist extension (4), radial deviations (5), ulnar deviations (6), extended palm (7), each performed for 3 seconds with a 3-second pause between gestures. The *extended palm gesture* class was excluded from the training phase due to its unavailability for multiple subjects.
- 4) Lastly, the BandMyo dataset includes signals recorded from 8 sensors using a Myo armband. It encompasses 15 distinct hand gestures, categorized into five finger gestures, six wrist gestures, and four other hand motion gestures, with each gesture repeated 8 times. Data collection involved 6 intact subjects, who removed the armband and took inter-repetition breaks, thereby introducing greater signal variance between repetitions.

Our aim in incorporating these diverse EMG datasets is to demonstrate the broad applicability of our GCN classifier through extensive experiments. Each dataset exhibits unique attributes, including variations in the number of subjects, the diversity and configurations of EMG sensors utilized, as well as disparities in data collection and preprocessing methodologies, and the range of gesture classes represented. These distinctions afford us a rich landscape for evaluating the robustness and generalization capabilities of our classifiers. A comprehensive overview of the datasets is provided in Table I, outlining key characteristics and contextual understanding.

- 2) **Compared Methods:** We conduct a comprehensive comparison of our proposed method with state-of-the-art methods, including models based on CNNs (*i.e.*, Erözen et al. [26],

TABLE I
SUMMARY OF ALL DATASETS USED IN THIS STUDY

	BandMyo	Ninapro DB5	UCI	CapgMyo Type A	CapgMyo Type B
# Subjects	6	10	36	18	20
# Gestures	15	53	7	8	8
# Repeats	8	6	2	10	10
# Sensors	8	8+8	8	16×8	16×8
Sample Rate (Hz)	200	200	200	100	100
EMG Device	Myo	2 Myo	Myo	HD-sEMG	HD-sEMG
Gesture Duration	3-4 sec	5 sec	3 sec	1 sec	1 sec

Xu et al. [14], Pancholi et al. [20], Shen et al. [27], Wei et al. [21], Wei et al. [19], Wei et al. [22], and Bahador et al. [23]), models integrating CNNs and RNNs (*i.e.*, Karnam et al. [18], Rahimian et al. [35], Zhang et al. [15], and Li et al. [16], Hu et al. [17]), and methods based on GNNs (*i.e.*, Lai et al. [31] and Jia et al. [28]). These comparisons aim to highlight the superiority of our proposed method over existing state-of-the-art models in the standard intra-subject gesture classification setting.

Furthermore, models with six different graph representations (*i.e.*, KNN-1, KNN-1+SW, KNN-2, KNN-2+SW, FC, FC+SW) were evaluated and compared across all datasets. For the CapgMyo dataset, only KNN-1, KNN-1+SW, KNN-2, and KNN-2+SW were compared due to the computational cost for the construction of FC graph representation for this dataset. The comparison across these variants helps show the effectiveness of incorporating different graph structures into the GNN model and its impact on the overall model performance.

3) Training and Evaluation: We conduct intra-subject and inter-subject experiments on these five datasets. For all experiments, we randomly split the data into training and test sets for each subject with a ratio of 8 : 2, where the overall gesture classification accuracy is the average across all subjects in each database. All models were trained on V100 and A100 Tensorcore GPUs with 40GB VRAM. To accommodate the different characteristics of the datasets, the hyperparameter settings used in training are also different: for example, conventionally, the gesture segment's window sizes are set differently for each dataset: 200 ms for BandMyo, 400 ms for Ninapro DB5, 300 ms for UCI and 500 ms for CapgMyo. More details of the hyperparameter settings are presented in Table II.

B. Comparison With the State-of-the-Art

To demonstrate the effectiveness of our proposed method, we present quantitative results and comparison with the state-of-the-art methods in the intra-subject experimental setting. Table III presents comparative results of our proposed models with different graph structures and state-of-the-art models. As shown in the table, while prior studies utilizing CNN methods have improved significantly over the years, a majority

of the studies report their investigations on only one dataset. Across all datasets, existing models showcase the most promising results with CapgMyo due to its smaller number of gesture classes and the utilization of high-density sensor arrays. This configuration allows the extraction of dense features from the abundant sensors. However, on datasets with sparse sensor configurations and more gesture classes, such as BandMyo [15] and Ninapro DB5 [11], existing models struggle to effectively leverage the spatial relationships between diverse sensors, leading to suboptimal performances.

Overall, GNN-based methods, including ours, generally outperform CNN-based approaches in gesture classification [28], [31]. For example, Jia et al. [28] achieved accuracies of 79.3%, 99.8%, and 99.4% on BandMyo, CapgMyo Type A, and CapgMyo Type B, respectively. This advantage stems from the GNN's ability to explicitly model relationships between sensors through graph convolution layers, unlike CNNs which typically consider sensor measurements independently or use heuristic aggregation of neighboring channels. Our proposed method further enhances this capability by incorporating both spatial and temporal information within a single, unified graph structure. This leads to our best models (FC+SW and KNN-2+SW) achieving over 99.3% accuracy on all datasets, exceeding or matching state-of-the-art performance. This improvement can be attributed to several key factors:

First, our method captures both spatial proximity and temporal dynamics, allowing the model to learn relationships across muscle regions and time windows, which is crucial for understanding dynamic gestures. Conventional GNNs often focus solely on spatial relationships, limiting their ability to capture temporal information. Second, the combined spatiotemporal graph, coupled with an end-to-end GCN architecture, facilitates more effective learning of discriminative features for gesture classification. By considering both spatial and temporal context, the model can learn more nuanced representations compared to methods that process spatial and temporal information separately. Further, our approach not only improves performance but also offers better interpretability. The learned graph structure and associated weights provide insights into the importance of different spatial and temporal relationships for specific gestures, offering a more transparent understanding of the model's decision-making process. This is a significant advantage over black-box CNN approaches. Finally, the flexible spatiotemporal graph configuration allows the model to focus on the most relevant features for gesture recognition, potentially leading to greater efficiency compared to methods that rely on less informed feature selection. This adaptability also makes our approach robust across various datasets and sensor configurations. The following experiments will provide a systematic evaluation of these factors.

C. Comparison Across Different Graph Structures

The results in Table III also show that the selection of graph structures impacts the process of aggregating information carried out by the graph convolution layers, consequently influencing the feature representations and classification accuracy. The models with the FC+SW graph structure were particularly effective in bridging the gap in learning representative

TABLE II
HYPERPARAMETER SETTINGS USED IN THE EXPERIMENTS

Hyperparameter	BandMyo	Ninapro DB5	UCI	CapgMyo Type A	CapgMyo Type B
Window Size (ms)	300	400	300	500	500
Epochs	100	150	150	100	100
Dropout Rate	0.2	0.2	0.1	0.15	0.15
Batch Size	64	128	64	128	128
FC Layer Size	128	256	256	64	64
Learning Rate	Started 0.01, decayed 0.002 per 20 epochs				
Optimizer	Adam				

TABLE III

QUANTITATIVE RESULTS OF MODEL PERFORMANCES, WHERE PARENTHESES INDICATE THE UTILIZATION OF SUBSETS OF GESTURE CLASSES IN THE COMPARED STUDIES, WITH THE ACCOMPANYING NUMBERS DENOTING THE CLASS COUNT

Method	Network	BandMyo	Ninapro DB5	UCI	CapgMyo Type A	CapgMyo Type B
Erözen <i>et al.</i> [26]	CNN	–	–	83.9 (6)	–	–
Karnam <i>et al.</i> [18]	CNN-RNN	–	–	98.3 (6)	–	–
Xu <i>et al.</i> [14]	CNN	–	71.7% (18)	–	–	–
Pancholi <i>et al.</i> [20]	CNN	–	98.3% (18)	–	–	–
Rahimian <i>et al.</i> [35]	CNN-RNN	–	67.7% (52)	–	–	–
Shen <i>et al.</i> [27]	CNN	–	74.5% (52)	–	–	–
Zhang <i>et al.</i> [15]	CNN-RNN	71.7%	–	–	91.7%	90.3%
Li <i>et al.</i> [16]	CNN-RNN	–	–	–	96.6%	–
Wei <i>et al.</i> [21]	CNN	–	88.9% (41)	–	99.5%	–
Wei <i>et al.</i> [19]	CNN	–	90.1% (41)	–	–	–
Wei <i>et al.</i> [22]	CNN	–	–	–	99.7%	–
Hu <i>et al.</i> [17]	CNN-RNN	–	–	–	99.7%	–
Lai <i>et al.</i> [31]	GNN	75.8%	–	–	99.8%	99.4%
Jia <i>et al.</i> [28]	GNN	79.3%	–	–	99.8%	99.4%
Bahador <i>et al.</i> [23]	CNN	–	–	–	99.9%	–
KNN-1 (Ours)	GNN	76.8%	95.6%	91.3%	92.2%	76.7%
KNN-1+SW (Ours)	GNN	97.3%	97.1%	98.1%	97.8%	89.1%
KNN-2 (Ours)	GNN	94.6%	98.8%	98.2%	99.3%	99.2%
KNN-2+SW (Ours)	GNN	99.6%	99.3%	99.6%	99.9%	99.8%
FC (Ours)	GNN	98.1%	98.9%	96.3%	–	–
FC+SW (Ours)	GNN	99.7%	99.4%	99.9%	–	–

spatiotemporal features that could not be achieved through convolution. This results in the best performances across three of the five datasets.

Comparing across different spatial graphs, we observe that models trained with sparse connections, such as KNN-1, exhibit inferior performance due to the limited connectivity between graph nodes. Conversely, models trained with denser connections, such as KNN-2 and FC, achieve better performances as they are able to more effectively propagate relevant information across sensors. This finding underscores that increased connectivities across graph nodes are beneficial for the models to learn the spatial relationships across EMG signals derived from different sensors.

Furthermore, by incorporating sensor-specific connections, the models can effectively capture and explore the temporal characteristics of EMG signals. The results in Table III demonstrate that SW connections lead to a significant improvement in model performance. For example, the KNN-1 graph model trained on the BandMyo dataset had a 17.6% performance

increase (from 76.8% to 94.4%) when SW connections were added. This improved performance can be attributed to the fact that the addition of SW connections enables most graph structures to acquire representative spatiotemporal features that existing CNN and GNN methods struggle to achieve. This highlights the importance of incorporating temporal information in the learning process and its important role in training models to capture effective temporal representations.

Overall, these findings suggest that the dense spatial connections (KNN-2 or FC) and sensor-wise temporal connections complement each other, and the combination of both is an effective approach for learning representative spatiotemporal features for gesture classification.

D. Comparison of Inter-Subject Classification Results

In addition to achieving better intra-subject accuracy across various datasets, our models demonstrate high accuracy in inter-subject gesture classification. It is widely acknowledged that muscular activations and their corresponding statistical

TABLE IV
INTER-SUBJECT CLASSIFICATION RESULTS OF THE
PROPOSED METHOD

Method	BandMyo	Ninapro DB5	UCI	CapgMyo Type A	CapgMyo Type B
KNN-1	52.3%	68.7%	77.5%	68.2%	62.4%
+SW	69.4%	85.4%	86.7%	76.6%	86.4%
KNN-2	76.5%	83.4%	90.1%	84.7%	83.9%
+SW	96.3%	90.2%	94.3%	92.3%	93.7%
FC	77.1%	87.3%	92.5%	–	–
+SW	99.4%	92.4%	98.9%	–	–

patterns inherently differ among subjects. Therefore, the capacity for inter-subject generalizability holds paramount importance in the context of practical applications. Table IV presents the results of our models in inter-subject gesture classification. The observed classification results exhibit resemblances to the patterns observed in intra-subject classification results, where the level of graph connectivity correlates with the accuracy of gesture classification. In particular, we observe that the accuracy improves significantly from the KNN-1 model to the KNN-2 model (e.g., from 52.3% to 76.5%, a 24.2% increase on the BandMyo dataset), due to the increased spatial connectivity. In addition, the temporal SW connections have yielded a notable enhancement in model performance. For instance, the application of these connections has led to a substantial improvement in the KNN-2 model's accuracy (e.g., from 76.5% to 96.3%, a 19.8% increase on the BandMyo dataset).

These results demonstrate the effectiveness of our spatiotemporal graph approach for cross-subject generalization. By explicitly modeling both spatial and temporal dependencies, our GCN model can learn consistent EMG patterns across subjects, despite variations in muscular activations. Specifically, the spatial connections capture consistent spatial relationships between muscle groups involved in the same gesture, even if the precise activation levels vary between individuals; the temporal connections model the temporal dynamics of muscle activations, which are crucial for distinguishing gestures and are often preserved across subjects even when the magnitudes of activations differ. This combined spatiotemporal approach enhances generalizability, resulting in a more robust and adaptable model for real-world inter-subject gesture recognition scenarios.

E. Generalization to Other GNN Models

To further verify the generalizability of our proposed graph structures, we conduct experiments by substituting our GNN classifier with four prominent GNN models: graph recurrent neural network (GraphLSTM) [67], graph attention network (GAT) [68], spatiotemporal graph convolution network (STGCN) [69], and graph convolution LSTM network (GCLSTM) [70]. The experiments were implemented under intra-subject settings, employing the identical six graph structures, where the results of these four models are summarized in Table V. The performance outcomes across various

TABLE V
GESTURE CLASSIFICATION RESULTS OF THE FOUR GNN MODELS.
THESE RESULTS SHOWCASE THE GENERALIZABILITY OF OUR GRAPH
TOPOLOGIES ACROSS FIVE GNN-BASED MODELS (INCLUDING GCN)
WITHIN FIVE PUBLIC EMG-GESTURE DATASETS

Method	BandMyo	Ninapro DB5	UCI	CapgMyo Type A	CapgMyo Type B
Graph Recurrent Neural Network (GraphLSTM) [67]					
KNN-1	10.8%	20.2%	68.3%	90.6%	96.2%
+SW	31.0%	20.9%	79.0%	91.9%	97.5%
KNN-2	68.1%	20.9%	91.4%	91.6%	96.4%
+SW	72.8%	23.6%	95.5%	91.8%	97.3%
FC	77.4%	24.7%	99.9%	–	–
+SW	84.1%	26.6 %	99.9%	–	–
Graph Attention Network [68]					
KNN-1	45.4%	96.4%	74.8%	99.8%	99.9%
+SW	92.1%	97.3%	85.2%	99.9%	99.8%
KNN-2	99.2%	96.6%	98.6%	99.8%	99.9%
+SW	99.7%	98.2%	99.8%	99.9%	99.9%
FC	88.4%	98.3%	71.6%	–	–
+SW	99.3%	98.8%	97.7%	–	–
Spatio-Temporal Graph Convolution Network [69]					
KNN-1	51.5%	49.4%	65.1%	95.0%	97.9%
+SW	63.9%	50.2%	70.4%	97.1%	98.3%
KNN-2	93.7%	50.1%	73.9%	95.6%	98.4%
+SW	99.8%	50.5%	75.8%	97.7%	99.3%
FC	94.1%	51.3%	90.0%	–	–
+SW	99.8%	51.5%	88.4%	–	–
Graph Convolution LSTM Network [70]					
KNN-1	18.4%	47.2%	63.1%	95.1%	97.4%
+SW	66.9%	50.9%	66.1%	97.4%	97.6%
KNN-2	32.4%	52.4%	65.9%	97.1%	97.6%
+SW	96.4%	51.1%	67.2%	98.0%	98.5%
FC	85.0%	51.9%	85.9%	–	–
+SW	99.7%	54.6%	92.4%	–	–

datasets exhibit nuanced variations where the BandMyo and CapgMyo A/B datasets manifest superior performance with GAT, STGCN, and GCLSTM, achieving accuracies exceeding 98%. Conversely, in the UCI dataset, GraphLSTM and GAT models outperform their counterparts, nearly achieving 100% accuracy. However, the Ninapro datasets encounter challenges in achieving comparable performance levels attributed to the intricacies arising from a large number of gesture classes, with the exception of the GAT model. A salient observation emerges from the comparative analysis across the four models, wherein the classification accuracies of the GAT model demonstrate parity with the results delineated in Table III, with the KNN-2+SW model performing the best on four out of five datasets. In scenarios characterized by denser graph structures, augmented connectivity could potentially impede the efficacy of attention mechanisms by precisely concentrating on the most important features. Thus, the enhancement in performance across diverse models does

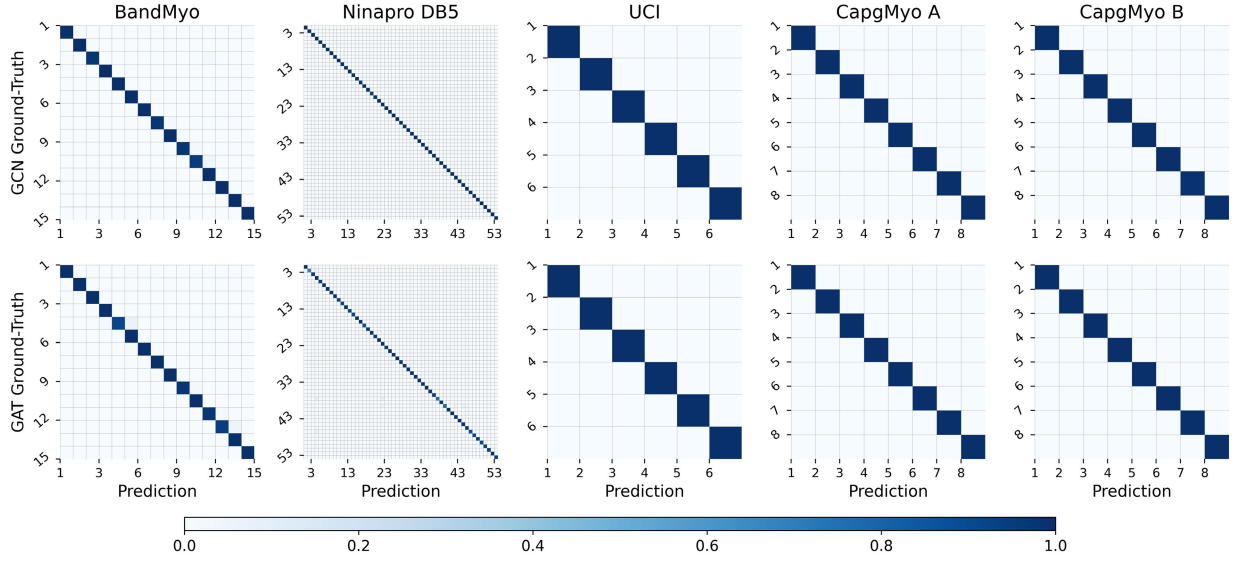


Fig. 6. The presented confusion matrices are shown conducted on five distinct datasets, employing both FC+SW graph architectures within GCN and GAT models. Notably, the CapgMyo datasets were subjected to a KNN2 augmentation with a SW structure. The depicted confusion matrices delineate intra-subject outcomes derived from the initial subject within each dataset. These outcomes, indicative of high accuracy, substantiate the efficacy of the proposed methodologies across a diverse array of publicly available databases, encompassing various gesture classes.

not consistently correlate with heightened graph connectivity. Instances occasionally arise where the KNN-2+SW or FC structure either outperforms or exhibits commensurate performance vis-à-vis the FC+SW configuration. This underscores the efficacy of sparse graph configurations in regulating information aggregation, thereby mitigating overfitting and facilitating the attainment of optimal performances through effective information propagation within the graph structure. Furthermore, given the high computational costs associated with the FC+SW configuration, particularly in scenarios featuring dense graph inputs such as in HD-sEMG settings, leveraging a cost-effective structure confers a significant advantage in enhancing practical feasibility by curbing computational overheads. To this end, we elucidate additional experiments aimed at optimizing resource utilization to achieve a balance between performance and computational efficiency in the subsequent subsection IV-G.

Additionally, we present visual representations of the confusion matrices in Fig. 6, showcasing the outcomes of our two top-performing models: the GCN and GAT. These models were trained utilizing the FC+SW graph structure, except for the CapgMyo datasets, which were trained with the KNN2+SW structure. These graphical depictions offer insights into the intra-subject performance outcomes across all datasets. Examination of these confusion matrices reveals consistently high accuracy along the diagonal segments across all datasets and their corresponding gestures. However, notable misclassifications are discerned in the GAT model's performance on the Ninapro DB5 dataset. Particularly, gestures such as 'Index flexion' and 'Quadpod grasp' exhibit a propensity for misclassification, while the gesture 'Prismatic four fingers grasp' is susceptible to being erroneously classified as 'Abduction of extended fingers' or 'Middle flexion'. These observations furnish insights into the performance

characteristics of the models and identify potential avenues for refinement. Observing these results, our approach demonstrates generalizability not only across datasets but also across different network architectures, underlining its robustness and adaptability.

F. Interpreting Spatial EMG Patterns

Prior studies have developed deep learning methods to interpret the biological relevance of hand movements from sEMG sensory data [71], [72], [73], [74], [75]. In our work, the graph structure takes into account the semantic relationships inherent in the spatial positions of EMG sensors, which contributes to a higher level of interpretability of the GNN models. In Section III-D, we have proposed a model interpretation method to address the challenge of understanding EMG-based classification models. This enabled us to assess the significance of specific EMG sensors that the model prioritizes. Fig. 7 illustrates an example of EMG array importance derived from the KNN-2+SW model trained on the CapgMyo Type A dataset, covering gesture classes 1 to 8 for subjects 1 to 6. Each vertical bar in the figure aggregates the importance (*i.e.*, the $I_{(\exists s, \exists g)}$ computed in Equation (13)) of an array of 8 sensors. As shown in the figure, the importance of sensor arrays remains consistent within individual subjects, while inter-subject quantification exhibits heterogeneity even for identical gesture classes. This underscores the notable variance in muscular activation extent across subjects. Notwithstanding inter-subject variances, it is noteworthy that the distinct models trained for individual subjects consistently assign higher priority to sensor arrays 10 and 16 for the majority of gestures. This observation remains consistent across various graph structure results trained with the CapgMyo Type A dataset, indicating the critical importance of the muscular regions in which sensor arrays 10 and 16 are situated. In summary, this figure

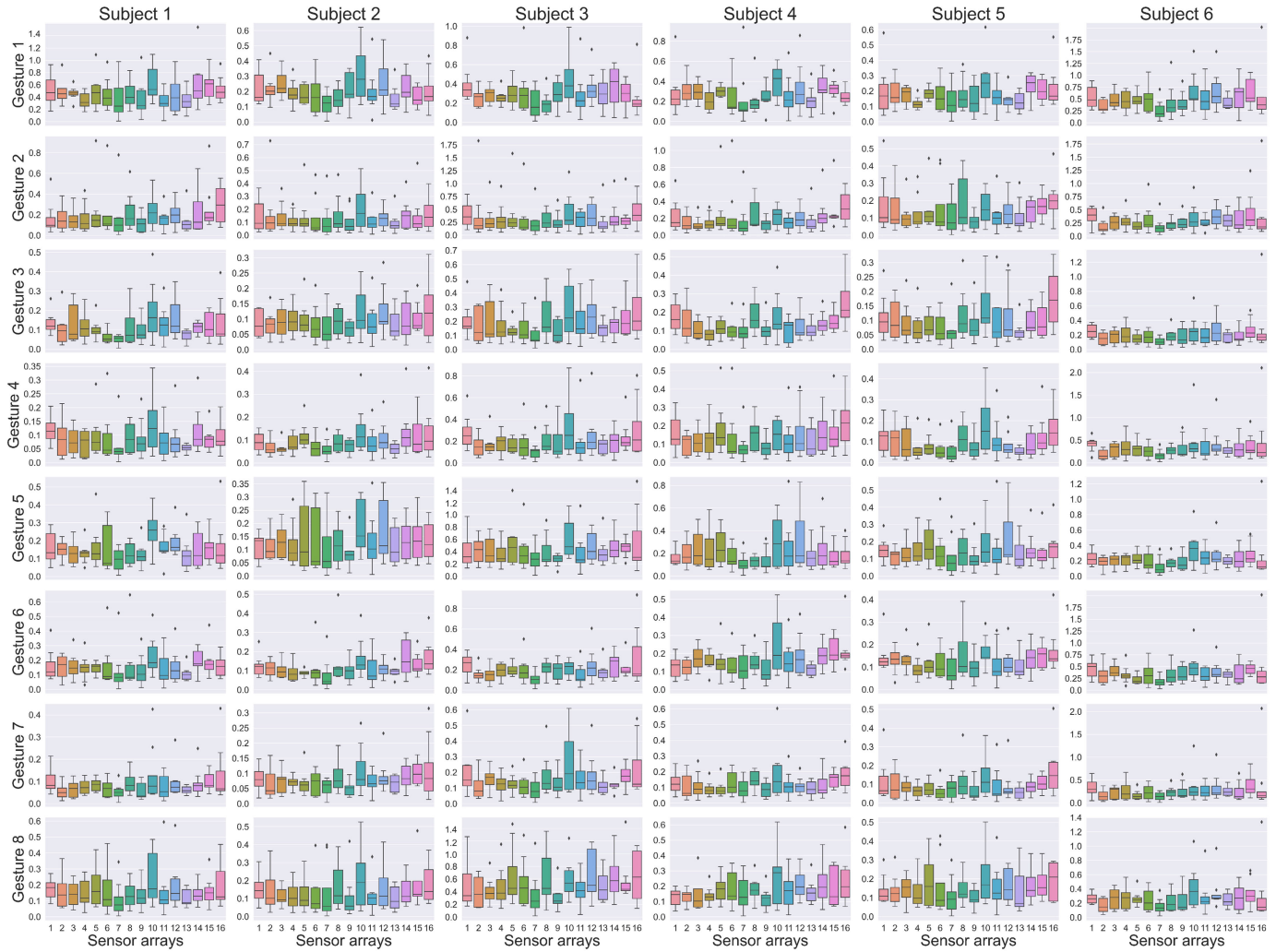


Fig. 7. The importance of EMG sensor arrays in classifying different gestures. The sensor importance scores are computed based on the proposed GCN model with the KNN2+SW graph structure trained on the CapgMyo Type A dataset.

shows how our method excels in providing clear insights into the specific muscle areas that the models prioritize when classifying gestures, a significant improvement over previous opaque neural network approaches.

However, it is noteworthy that the interpretation results varied across models. For instance, although the GAT demonstrated comparable performance, its interpretability results diverged significantly. This variability, despite being trained and tested on the same datasets, can be attributed to differences in the models' underlying architectures and training mechanisms. While most model algorithms aim to minimize loss using distinct loss functions, their sample-level predictions often differ. For example, one model might be particularly adept at distinguishing certain gesture types, showing sensitivity to specific data distributions, while another may not exhibit the same behavior. As a result, differences in sensor activation and other interpretive outputs are unavoidable, reflecting the natural consequence of the diverse algorithms employed.

G. Optimization of Sensor Configuration

Despite leading in gesture recognition, the GNN model's computational demands restrict its practical deployment in

real-world applications [76], especially in HD-sEMG scenarios. For example, using all 128 sensors of the CapgMyo configuration leads to computationally intensive models. To lower the computational cost, we aim to minimize sensor use while retaining classification performance. We perform two experiments on CapgMyo type A and B datasets: 1) randomly reducing sensors and 2) utilizing each row of sensors. These approaches evaluate the impacts of both sensor quantity and position, offering valuable insights for informed sensor selection decisions.

In our first experiment, we evaluate the classification performance by varying the number of engaged sensors. Following the experiment protocol of the K-stratified random forest (KSFR) approach [77], we apply our GCN model with the KNN2+SW structure to randomly sampled sensor subsets, varying from 28 to 128 sensors in increments of 25 sensors. As shown Fig. 8, with only 28 sensors, our model outperforms KSFR significantly, attaining an accuracy of 96.8% on the Type A dataset and 97.1% on the Type B dataset. Compared to the full model with 128 sensors, the accuracy gap is less than 3.1%. These results represent an approximately 50% improvement over KSFR, highlighting the efficiency of our

TABLE VI
A COMPARATIVE PERFORMANCE ANALYSIS ON INDIVIDUAL ROWS (R1 – R8) WITHIN CAPGMYO SENSOR ARRAYS.
RESULTS ARE AVERAGED ACROSS ALL SUBJECTS

	CapgMyo Type A								CapgMyo Type B							
	R1	R2	R3	R4	R5	R6	R7	R8	R1	R2	R3	R4	R5	R6	R7	R8
GCN (KNN1)	96.1	96.6	96.2	97.2	96.9	96.5	96.6	95.5	86.1	88.3	88.2	95.7	94.4	88.8	85.5	85.1
GCN (KNN1+SW)	96.6	97.3	97.4	97.5	97.3	97.1	97.5	96.8	87.4	87.3	88.8	96.2	96.0	89.2	88.6	87.5
GCN (KNN2)	97.2	96.1	96.7	97.2	97.7	96.7	97.1	96.2	88.3	87.9	88.2	95.7	94.5	89.1	86.1	88.1
GCN (KNN2+SW)	97.8	96.5	97.9	98.0	97.9	97.8	97.2	96.5	90.6	88.5	90.0	97.0	96.4	91.6	88.5	89.2

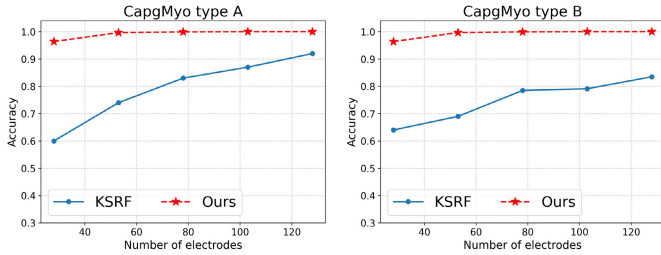


Fig. 8. Results with reduced sensor configurations on the CapgMyo dataset.

method in maintaining competitive performance under reduced sensor configuration. The superior performance of our method can be attributed to its consideration of the spatial configuration of sensors as well as temporal relationships, taking into account relevant biological indications from the HD-sEMG data, such as muscle activity patterns, spatial distribution, and temporal aspects of muscle activation, which ensures that the model aligns well with the physiological aspects of gesture production.

Our next experiment focuses on the positioning of sensors by utilizing a single row of sensors from the CapgMyo HD-sEMG array. Each row comprises 16 sensors aligning vertically with forearm muscles. Table VI compares the classification performance based on each row of sensory data (*i.e.*, R1 – R8). It shows that for all GCN with graph structures, the R4 and R5 sensors exhibit consistently superior performance on both Type A and Type B datasets, suggesting that the sensory data in the 4th and 5th rows holds critical information for accurate hand gesture classification. The likely factors contributing to their high performance include their positioning relative to forearm anatomy, consistent signal quality, broad spatial coverage of forearm muscles, and proximity to nerve clusters associated with hand movements. The results show that our GCN with graph structures effectively simulate muscular activities by learning information transmission paths, creating denser connectivity, cycles, and parallel propagation for efficient gesture classification [78], [79]. These findings emphasize the importance of sensor placement and offer insights into optimizing sensor array configurations for accurate hand gesture recognition.

V. CONCLUSION

In conclusion, our research represents a significant advancement in EMG-based gesture classification by leveraging novel graph-based representations and graph convolution network (GCN) model. Through the incorporation of prior anatomical and physiological knowledge of the upper limb and

their sensory configuration, our approach effectively captures spatial and temporal relationships among EMG sensors and signals, leading to improved efficiency, generalizability, and interpretability of gesture recognition systems. We have demonstrated the efficacy of our approach through comprehensive experimental evaluations, showcasing state-of-the-art performance across multiple public EMG datasets. Notably, our classifiers exhibit robustness and adaptability across diverse datasets, subjects, and network architectures, surpassing other existing methods in terms of generalizability. Furthermore, our experiments extend to performance optimization with reduced sensors, where we achieve comparable accuracy while significantly reducing the number of sensors used in high-density sEMG configurations. This breakthrough not only enhances the practical feasibility of applying graph models in the sEMG domain but also underscores the efficiency gains of our approach. Importantly, our research offers valuable insights into the biological mechanisms underlying human upper-limb movements. The interpretability of our graph-based approach reveals consistent sensor activation patterns correlated with distinct muscle areas for recognizing various hand motion gestures. This interpretability aspect represents a significant advancement in EMG-based hand gesture classification, enhancing the effectiveness, adaptability, and comprehension of gesture recognition systems.

To summarize, our research presents a promising avenue for advancing gesture recognition technology, with wide-ranging implications across real-world biomedical and human-to-machine applications. By merging graphical models built with insights from human anatomy and physiology, we have constructed a robust upper-limb gesture recognition framework for accurate, generalizable, and interpretable gesture classification. We believe that our approach holds large potential for revolutionizing the development of clinically practical systems capable of seamlessly adapting to new subjects and sensor configurations, thereby improving the quality of human-computer interaction and engineering-based rehabilitation systems across diverse domains.

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